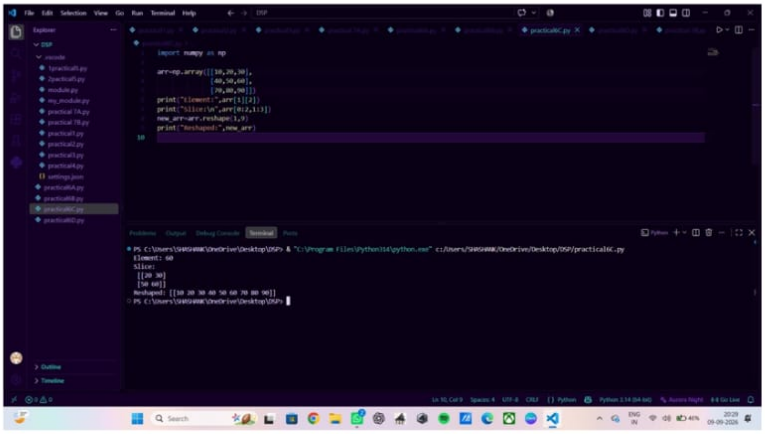
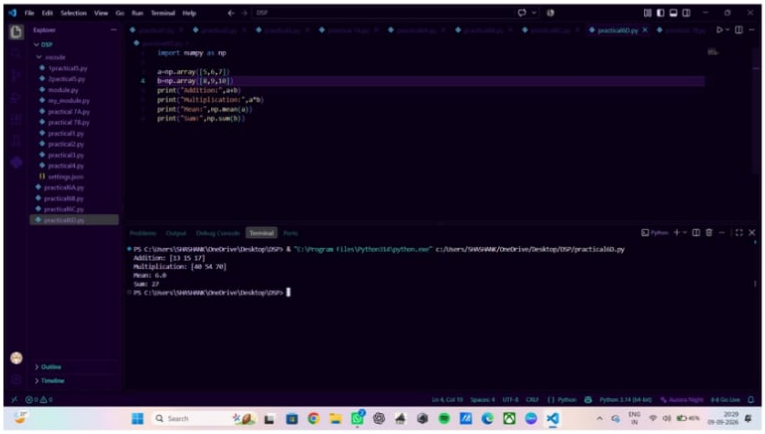


```
import numpy as np

arr=np.array([[10,20,30],
              [40,50,60],
              [70,80,90]])
print("Element:",arr[1][2])
print("Slice:\n",arr[0:2,1:3])
new_arr=arr.reshape(1,9)
print("Reshaped:",new_arr)
```

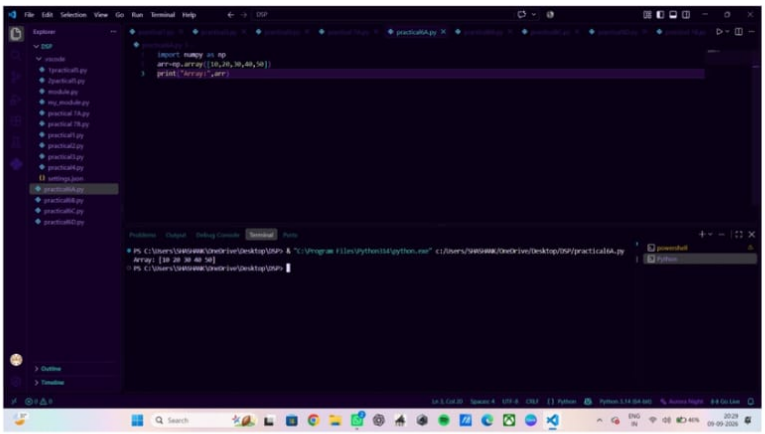
```
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP> & "C:\Program Files\Python314\python.exe" c:/Users/SHRSHANK/OneDrive/Desktop/DSP/practical6C.py
Element: 60
Slice:
 [[20 30]
 [50 60]]
Reshaped: [[10 20 30 40 50 60 70 80 90]]
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP>
```



```
import numpy as np

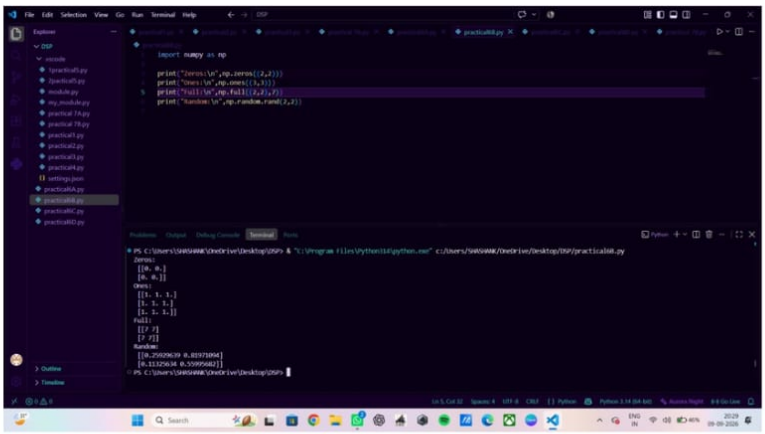
a=np.array([5,6,7])
b=np.array([4,9,10])
print("Addition:",a+b)
print("Multiplication:",a*b)
print("Mean:",np.mean(a))
print("Sum:",np.sum(b))
```

```
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP> & "C:\Program Files\Python314\python.exe" c:/Users/SHRSHANK/OneDrive/Desktop/DSP/practical6D.py
Addition: [13 15 17]
Multiplication: [40 54 70]
Mean: 6.0
Sum: 27
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP>
```



```
import numpy as np
arr=np.array([10,20,30,40,50])
print("Array:",arr)
```

```
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP> & "C:\Program Files\Python314\python.exe" c:/Users/SHRSHANK/OneDrive/Desktop/DSP/practical6A.py
Array: [10 20 30 40 50]
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP>
```



```
import numpy as np

print("Zeros:\n",np.zeros((2,2)))
print("Ones:\n",np.ones((3,3)))
print("Full:\n",np.full((2,2),7))
print("Random:\n",np.random.rand(2,2))
```

```
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP> & "C:\Program Files\Python314\python.exe" c:/Users/SHRSHANK/OneDrive/Desktop/DSP/practical6B.py
Zeros:
 [[0. 0.]
 [0. 0.]]
Ones:
 [[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
Full:
 [[7 7]
 [7 7]]
Random:
 [[0.25929639 0.81973894]
 [0.11329634 0.53995602]]
PS C:\Users\SHRSHANK\OneDrive\Desktop\DSP>
```